The Psychology of E-Learning

Caught in the Web: Internet Abuse

Research on Internet Abuse

- Internet use
  - The positive side: facilitate the flow of **information worldwide**.
  - The negative side:
    - Many people feel **helpless** and **out of control**, and report job and school impairments, interpersonal problems.
    - Parents and teachers **distress** by **loss of the ability to monitor** children's internet use.

Research on Internet Abuse

- Definition for internet abuse, internet addition, internet dependency, and pathological internet use:
  - "The common denominator is that individual appears to others to have **diminished control** over the behaviors or report a sense of **disequilibrium**, **loss, and distress** when the object or substance is unavailable or the behaviors are curtailed" (Jaffe, 1990).

Research on Internet Abuse

- Assessment of internet abuse
  - **No studies** have defined internet abuse **only** by the **amount of time** spent on-line.
  - The criteria have included tolerance and continued used despite recurrent internet-related problems.

Research on Internet Abuse

- Incidents of internet abuse
  - The incidents of IA among college and high school students in the United States, Europe, and Taiwan consistently has been in the **5.9-13%.**
  - In Taiwan, the incidence of IA was **10.9%** in the sample of 616 **high school students**, and **5.9%** in a study of 12 **universities**.

Research on Internet Abuse

- On-line versus off-line studies
  - What are the differences ?
  - **A bias that along with the sample to those who use the internet more.**

Research on Internet Abuse

- On-line versus offline studies
  - **74 to 80%** on-line participants met the criteria for IA, while non-self-selected off-line studies varied from **8 to 13%,** reflecting a **self-selection bias**.
  - The **on-line** participants report **more internet-related problems** than offline studies (**46% Vs. 3%).**

Research on Internet Abuse

- Time on-line
  - Time spend on the internet was **significantly higher** for studies done **on-line**.

Research on Internet Abuse

- Demographics (gender)
  - In general population, **males spend more time** on-line than females.
  - Males with IA were more than females in some studies
    - **2-5 times more** than females in off-line studies.
    - 17% Vs. 8%; 12.2% Vs. 3.2%; 15.8% Vs.2.7%; 12% Vs. 5.3%
    - Why???

Research on Internet Abuse

- Demographics (gender)
  - Some other on-line studies that reported no gender difference or that **females** are more likely to be IA.
    - Why females?
      - **Methods** used to recruit participants
      - Females are more likely to **seek help for problems.**
      - Females are likely to **participate in psychological studies**
      - Females are more likely to **respond for IA**.

Research on Internet Abuse

- Demographics (age)
  - The incidence of IA is **higher** among **teens and young adults** (Why?)
    - Heaviest users of the internet
    - Particularly sites used by those develop IA

Research on Internet Abuse

- Demographics (age)
  - Studies have found
    - **Teenagers** most **vulnerable** to the potential **negative** effects of the internet.
    - **Younger** participants experienced **more problems** from internet use than older individuals.
    - **Younger** people at **increased risk** of being **internet addicts**.

Research on Internet Abuse

- Ways of using the internet
  - IAs are more likely to use **socially interactive** aspects of the internet.
    - Chatrooms, discussion groups, and bulletin boards systems (BBS),
    - MUDs provides a recreational outlet and provide a place to meet people
    - E-mail: Some studies found that IAs use more often than others, but others didn't.
    - Facebook ,Line ,TikTok &Instagram (IG)

Research on Internet Abuse

- Ways of using the internet
  - Gambling on-line or cybersex
  - **No differences** found in the information gathering or using the internet for work or school

Explanations for Internet Abuse

- Changes in social interaction
  - Internet communication (e.g. chat rooms) is different from real life. (Why?)
    - Text-based anonymous
    - From person to person to “many to many”
    - Broaden social contacts
    - Available for 24 hours
    - Written text could reflect deeper thoughts

Explanations for Internet Abuse

- Changes in social interaction
  - The **lack of face to face** (ftf) communication and **anonymity** on-line can **lessen social risk**.
  - People can **control social distance** from others, thus providing controlled intimacy.
  - People can **express controversial** or **hidden thoughts** and feelings **without the fear** of RL rejection or confrontation.
  - On-line communities sometimes provide a **strong sense of acceptance.**

Explanations for Internet Abuse

- Changes in social interaction
  - Non IAs use the internet to gather information or maintain existing friendships, while **IAs** use the internet **primarily to meet** and **socialize** with **new people**. (why for IAs?)
    - Feel lonely
    - Experience more intense feeling of intimacy on-line that in ftf relationship than others (75% Vs. 38%)

Explanations for Internet Abuse

- Changes in social interaction
  - Findings for Morahan-Martin and Schumacher's research
    - **IAs (or PIU)** reported that **internet is easier** for them to **communicate** than others (91% Vs. 56%).
    - **IAs** reported that **most** of their friends were **on-line friends**

Explanations for Internet Abuse

- Changes in social interaction
  - Findings for Morahan-Martin and Schumacher's research
    - **IAs** were more than **twice** as others to **pretend** someone they are not, and gender swap on-line.
    - **IAs** felt that internet communication is **more liberating**, which enhanced their social behaviors.

Explanations for Internet Abuse

- Escape, relaxation and excitement
  - **IAs** were more likely to use the internet for **recreation**, **relaxation**, **wasting time** and **escaping pressure**.
    - **50% IAs** Vs. **25%** (limited) Vs. **18%** (regular) to **escape pressure**
    - Two **Taiwanese** studies showed that **IAs** went on-line for **escape**, **fun**, and **excitement.**
    - The use of **sexual** outlets and **gambling** were associated with a **higher** incidence of **IA**.

Explanations for Internet Abuse

- Mood alteration
  - **IAs** were more likely to go **on-line** to **escape negative emotions.**
    - IAs were more likely to use the internet to **escape pressure when down, anxious or isolated**.
  - On-line environments provide **alternate realities**, such as perceptions in **time**, and loss of **boundaries**, which facilitate mood alterations.

Explanations for Internet Abuse

- Power and mastery
  - People attempt to achieve a **sense of power** and **mastery on-line**, especially if they **cannot** achieve mastery and power in the **real world**.
    - A sense of **power** can be derived from users' ability to **master the virtual environment**.

Explanations for Internet Abuse

- Other explanations
  - **IAs** were **less likely** to use **task-oriented styles** and **more** likely to use **emotional coping styles** and to have a **more external locus of control** than others.
  - Some studies found that **higher incidence of IA** among college students in **technological, science**, and object-oriented fields, but as the **internet became easy to use**, the **technological** sophistication should **diminish** as a factor.

Internet Abuse: Symptom, Cause or Effect

- Cause and Effect
  - There is **no consensus** about whether there is a cause and effect relationship between internet abuse and other disorders.
  - The relationship may be **bidirectional.**

    - For those with other addictions, **internet** provide a **new channel** to **act out their pathologies** (e.g. gambling) and vice versa.

Internet Abuse: Symptom, Cause or Effect

- Cause and Effect
  - The cause and effect **relationship** among **loneliness, depression**, and the higher incidence of **IAs** is **uncertain**.
    - People who are **lonely or depressed** may turn to the internet, and may find **companionship, or social support.**
    - People **increasingly rely on on-line support** at the **expense of fostering RL contacts and support**.

Internet Abuse: Symptom, Cause or Effect

- Cause and Effect
  - The results of one longitudinal study support that **internet use causes**.
    - Both **loneliness and depression increased** with **greater internet use** on-line.
    - The researchers attribute the **increases in loneliness and depression** to **decreases in family communication, and social activities**.

Internet Abuse: Symptom, Cause or Effect

- Cause and Effect
  - Some disorders may be **coexist**.
    - Depression and loneliness were frequently coexist with a number of other disorders, such as eating disorders, anxiety disorders and personality disorders.

Internet Abuse: Symptom, Cause or Effect

- Alternative explanations
  - Research suggest that **internet abuse** may reflect **fears about a new form of technology.**
  - The **rapid expansion** of the internet combined with the perceived and **real sense of lack of control of access to information** were also **threats** for many people.

Internet Abuse: Symptom, Cause or Effect

- Alternative explanations
  - Although research suggests that the quality of on-line interactions and friendship were inferior than those in RL, but some argue that they could be **healthy social relationship**.
  - **On-line** communities can provide a **strong sense of community, companionship, acceptance** and **social support**, and many **on-line relationship were transferred into real life**.

Internet Abuse: Symptom, Cause or Effect

- Alternative explanations
  - **IA** is **not** just a **temporary phenomenon** among **new users**, research indicated that internet use is **higher among experienced** than inexperienced ones.
  - A study found that **heaviest internet use** with **new users**, especially **teens**, **peaks within the first few weeks** of being on-line.
  - Whether the IA is a temporary phenomenon is still **unknown.**

College Students and Internet Use and Abuse

- Developmental issues for college students
  - Identity
  - Separation
  - Autonomy
  - Intimate relationship
  - Balancing the social, academic and work-related demands of college

College Students and Internet Use and Abuse:

- Why the internet?
  - College students have much freedom from parental restrictions and from the rigid structure high school, so they have **much unstructured time**.
  - Internet environment provide the virtual world for college students to deal with their **developmental issues,** such as identity, isolation, intimacy and dating relationship.
  - Internet (interactions) also allow college students to **learn social skills** that generalize to their real life.

各國成癮比例

作者	年代	國家	調查時間	樣本對象(人)	年齡層	成癮比例 (%)
鄭照順、鄒浮安	2011	台灣	2010	1168	大學二年級	27.76%
Ko, Yen, Yen,Chen, & Chen	2012	台灣	2009	2162	11~13	10.8%
		台灣	2007	2114	國中生	17.9%
		台灣	2009	9405	12~16	18.8%
		台灣	2009	2453	大學生	12.3%
		土耳其	2009	292	14~19	11.6
		義大利	2009	98	14~19	36.7%
		希臘	2008	2200	12~18	8.2%
		挪威	2009	3399	16~74	1%

各國成癮比例

作者	年份	國家	調查時間	樣本對象 (人)	年齡層	成癮比例 (%)	工具
Ko, Wang, Liu, Yen, Chen, & Yen	2014	台灣	2005	2293	國中生	2.6%	Behavior Inhibition Scale, Behavior Approach Scale
Ha & Hwang	2014	韓國	2010	56086	國高中生	2.8% 2.4 % were at risk	Internet addiction self-scale short form (KS scale)
Tsitsika, Janikian, Schoenmakers, Tzavela, Ólafsson, Wójcik, ... & Richardson	2014	7個歐洲國家(西班牙、波蘭、德國、荷蘭、羅馬尼亞、冰島及希臘)	2011	13284	9年級與10年級	1.2% 12.7% at risk 38.9% mild signs	Internet Addiction Test（IAT ）
Adiele, & Olatokun	2014	南非	2013	1022	大學生	3.3% 20.3% were at risk	Internet Addiction Test (IAT)
Lopez-Fernandez, Honrubia-Serrano, Gibson & Griffiths	2014	英國	2013	1097	11-18	5.2% 14.2% were occasional 65.3% were habitual 15.3% were at risk	Internet usage, the problematic Internet use scale (PIEUSA)
國家發展委員會	2015	台灣	2015	1508	12-19	5%	The Chen Internet Addiction Scale(CIAS)
國家發展委員會	2017	台灣	2017	1508	18-19	5.9%	The Chen Internet Addiction Scale(CIAS)

各國成癮比例

作者	年份	國家	調查時間	樣本對象(人)	年齡層	成癮比例(%)	工具
Pan & Yen	2018	台灣	2018	1735	平均年齡 15.93 歲	23.0%	Chen Internet Addiction(CIAS)
Iernja, Vejmelka, & Rajter	2019	克羅地亞	2019	352	15-19 歲	39%低度成癮 32% 中度成癮 3.4%重度成癮	Internet Addiction Test(IAT)
Kentaro Kawabe, Fumie Horiuchi, Yasunori Oka, & Shu-ichi Ueno	2019	日本	2019	874	12 -15歲	23.7%	IAT
K. RACHUBInSKA, A. CYBULSKA, M. SZKUP, & E. GROCHANS	2021	波蘭	2021	556	平均年齡 34 歲	25%	IAT
Daisy Araújo Rodrigues, Inês Carvalho Relva, & Otília Monteiro Fernandes	2022	葡萄牙	2022	568	14 -18	73.1%	Internet Addiction Test (IAT)

全球各年齡層網路成癮比例

- 全球跨年齡層往成癮百分比
  - **Pew Research Center, 2019**

年份	年齡層	成癮比例
2019	10-19years	48%
	30-49 years	36%
	50-64 years	19%
	65+ years	7%

台灣各年齡層網路成癮比例

作者	年代	國家	樣本對象（人）	年齡層	成癮比例（%）	使用工具
國家發展委員會	2021	台灣	81	12-17 歲	9.6%	CIAS-10
			197	20-29 歲	11.2%	
			224	30-39 歲	9.1%	
			244	40-49 歲	8.6%	
			235	50-59 歲	6.9%	

網路成癮性別比例

- **成癮類項(含男VS.女)**
  本研究有效問卷共451份，其中男女比例接近1:1(46%:54%)(Young & Case, 2009)。

- Table1. Response Rate By Gender and Age (Young & Case, 2009)

Age	Male	Female
<20 years	29%	23%
20-29	33%	29%
30-39	14%	16%
40-49	14%	6%
50-59	5%	5%
60+	4%	2%
Blank	1%	19%
Total	100%	100%

台灣網路成癮性別比例

◆ 成癮比例(含男VS.女)

◆ 110年網路沉迷調查(國家發展委員會)

項目	有關網路成癮傾向的佔比		
	106年(%)	110年(%)	變動(百分點)
全體	5.5	7.0	+1.5
男	5.6	6.4	+0.8
女	5.4	7.6	+2.2

網路成癮活動性別比例

- 成癮類項(含男VS.女) 成癮比例
  - Table2. Internet Behaviors by Gender Issues in Information Systems (Young & Case, 2009)

Online Activity	Male	Female
Chat Rooms	22%	20%
Instant messaging	59%	57%
Pornography	58%	18%
Gambling	6%	4%
Gaming	51%	40%
Auction Houses	24%	22%
Shopping	24%	31%
Stock Watching	8%	5%
Download Music	52%	47%
Personal Email	78%	76%

網路活動類項之性別比例

- 常用網路活動類項(性別比例)
  - 本研究以國高中生為調查對象，有效問卷共25573份，高中生平均年齡17.2歲(男生佔50.9%)，國中生平均年齡14.32歲(男生佔48.9%)，男女比例接近1:1(Lee, Ko, & Chou, 2014)。
  - Frequencies and Percentage of the Most Frequent Online Activities (Lee, Ko, & Chou, 2014)

Online Activity	Junior High Boys (n = 6005)	Junior High Girls (n = 5770)
Making Friends	12.1 %	10.4 %
Online Gaming	43.3 %	29.1 %
School work	11.3 %	14.0 %
Entertainment	19.4 %	16.6 %
Video	22.2 %	24.1 %
Online Chatting	26.0 %	35.2 %
Blogging	15.1 %	32.7 %
Photo	8.8 %	22.0 %

網路活動類項之性別比例

- 常用網路活動類項(性別比例) (接續上頁)
  - Frequencies and Percentage of the Most Frequent Online Activities (Lee, Ko, & Chou, 2014)

Online Activity	Senior High Boys (n = 7037)	Senior High Girls (n = 6761)
Making Friends	11.4 %	7.5 %
Online Gaming	36.2 %	22.3 %
School work	16.0 %	19.1 %
Entertainment	22.8 %	20.1 %
Video	38.1 %	38.0 %
Online Chatting	24.4 %	24.2 %
Blogging	24.9 %	31.9 %
Photo	17.9 %	22.9 %

網路成癮活動性別比例

- 成癮類項(含男VS.女) 成癮比例
  - Table3.106年網路沉迷調查(國家發展委員會)

Online Activity	Male	Female
經營網路社群	15%	21%
瀏覽網路社群	15%	18%
使用聊天、通訊軟體	27%	38%
搜尋資料	28%	28%
看影片	29%	27%
遊戲	31%	18%
網路購物	24%	8%
聽音樂或廣播	5%	6%
看漫畫、小說	4%	4%
閱讀新聞	18%	16%

網路成癮活動性別比例

◆ 成癮類項(含男VS.女) 成癮比例(接續上頁)

Online Activity	Male	Female
查看股票金融訊息	3%	2%
搜尋色情相關資訊或影片	1%	2%
收發e-mail	1%	1%
學習	1%	1%
上網只有工作的目的	1%	2%
其它	1%	1%

台灣網路成癮比例

◆ 成癮類項(含有網路成癮傾向VS. 無網路成癮傾向) 成癮比例

◆ 110年網路沉迷調查(國家發展委員會)

上網活動	影片類活動	社交類活動	資訊類活動	遊戲	經濟類活動	其他娛樂	色情相關活動	無法分類
有網路成癮傾向	66.3%	47.8%	24.3%	22.9%	14.3%	5.8%	0.0%	0.0%
無網路成癮傾向	56.1%	48.8%	30.3%	19.0%	10.3%	8.1%	0.2%	0.9%

＊此問卷選項為複選題，百分比計算方式＝票數/總人數

台灣手遊成癮比例

◆ 成癮比例

◆ Development of the Problematic Mobile Gaming Questionnaire (PMGQ) and Prevalence of Mobile Gaming Addiction among Adolescents in Taiwan (Pan, Chiu & Lin , 2019)

年齡層	成癮比例	願意在遊戲中消費比例
小學生	19.1%	21.4%
中學生	20.5%	36.3%
高中生	19.0%	42.2%

台灣手遊成癮比例

- 「手遊成癮」：判斷成癮的主要因素包括「過度沉迷，產生負面影響」、「耐受性」、「戒斷症狀」，也與傳統上的「成癮行為」相當類似

樣本對象	比例
小學	19.1%
初中	20.5%
高中	19.0%

國家衛生研究院2020發布